

1. Analytical Derivation of Sphere-of-Influence Encounter Points

1.1. Problem Statement

Given two Keplerian elliptical orbits (spacecraft orbit 1, planet orbit 2) in heliocentric ecliptic coordinates, find the eccentric anomalies E_1, E_2 such that the distance between the spacecraft and the planet equals the planet's sphere-of-influence radius r_{SOI} :

$$\|A\mathbf{r}_1(E_1) - B\mathbf{r}_2(E_2)\| = r_{\text{SOI}} \quad (1)$$

where A and B are known constant 3×2 rotation matrices mapping from orbital-plane 2D coordinates to 3D heliocentric ecliptic coordinates.

1.2. Notation

- a_i, b_i, e_i : semi-major axis, semi-minor axis, and eccentricity of orbit i (with $b_i = a_i \sqrt{1 - e_i^2}$)
- E_i : eccentric anomaly of orbit i
- $A \in \mathbb{R}^{3 \times 2}, B \in \mathbb{R}^{3 \times 2}$: orbital-to-ecliptic rotation matrices with components A_{jk}, B_{jk}
- r_{SOI} : radius of the planet's sphere of influence

1.3. Position in Orbital Plane

For a Keplerian ellipse, the 2D position vector in the orbital plane as a function of eccentric anomaly is:

$$\mathbf{r}_i(E_i) = \begin{pmatrix} a_i(\cos E_i - e_i) \\ b_i \sin E_i \end{pmatrix} \quad (2)$$

1.4. Distance Constraint

Square both sides of the constraint:

$$(A\mathbf{r}_1 - B\mathbf{r}_2)^\top (A\mathbf{r}_1 - B\mathbf{r}_2) = r_{\text{SOI}}^2 \quad (3)$$

Expand the left-hand side:

$$\mathbf{r}_1^\top A^\top A \mathbf{r}_1 - \mathbf{r}_2^\top B^\top A \mathbf{r}_1 - \mathbf{r}_1^\top A^\top B \mathbf{r}_2 + \mathbf{r}_2^\top B^\top B \mathbf{r}_2 = r_{\text{SOI}}^2 \quad (4)$$

Each term on the left side is a scalar, and transpose of a scalar doesn't change it, and by also applying matrix product transposition property $(QW)^\top = W^\top Q^\top$, we get the following expression:

$$\mathbf{r}_1^\top A^\top A \mathbf{r}_1 - 2\mathbf{r}_1^\top A^\top B \mathbf{r}_2 + \mathbf{r}_2^\top B^\top B \mathbf{r}_2 = r_{\text{SOI}}^2 \quad (5)$$

1.4.1. Orthonormality of A and B

Since A and B are 3×2 matrices whose columns are orthonormal (they embed an orthonormal frame from the orbital plane into 3D space):

$$A^\top A = I_{2 \times 2}, \quad B^\top B = I_{2 \times 2} \quad (6)$$

Therefore the self-terms simplify:

$$\mathbf{r}_1^\top A^\top A \mathbf{r}_1 = \mathbf{r}_1^\top \mathbf{r}_1 = |\mathbf{r}_1|^2 =: r_1^2 \quad (7)$$

$$\mathbf{r}_2^\top B^\top B \mathbf{r}_2 = \mathbf{r}_2^\top \mathbf{r}_2 = |\mathbf{r}_2|^2 =: r_2^2 \quad (8)$$

1.4.2. The coupling matrix C

Define the 2×2 matrix:

$$C := 2A^\top B \quad (9)$$

This encodes the entire mutual orientation of the two orbital planes. Equation 5 becomes:

$$r_1^2 + r_2^2 - \mathbf{r}_1^\top C \mathbf{r}_2 = r_{\text{SOI}}^2 \quad (10)$$

Note: C is **not** necessarily symmetric — it is a general 2×2 real matrix with $|C_{jk}| \leq 2$.

1.5. Expanding the Constraint

1.5.1. Self-terms

Using $r_i = a_i(1 - e_i \cos E_i)$:

$$r_1^2 + r_2^2 = a_1^2(1 - e_1 \cos E_1)^2 + a_2^2(1 - e_2 \cos E_2)^2 \quad (11)$$

1.5.2. Cross-term

The position vector Equation 2 factors as $\mathbf{r}_i = \text{diag}(a_i, b_i)\mathbf{p}_i$, where:

$$\mathbf{p}_i = \begin{pmatrix} \cos E_i - e_i \\ \sin E_i \end{pmatrix} \quad (12)$$

Substituting into the cross-term:

$$\mathbf{r}_1^\top \mathbf{C} \mathbf{r}_2 = \mathbf{p}_1^\top \text{diag}(a_1, b_1) \mathbf{C} \text{diag}(a_2, b_2) \mathbf{p}_2 = \mathbf{p}_1^\top \mathbf{M} \mathbf{p}_2 \quad (13)$$

where the **scaled coupling matrix** is:

$$\mathbf{M} := \text{diag}(a_1, b_1) \mathbf{C} \text{diag}(a_2, b_2) = \begin{pmatrix} M_{11} & M_{12} \\ M_{21} & M_{22} \end{pmatrix} := \begin{pmatrix} C_{11}a_1a_2 & C_{12}a_1b_2 \\ C_{21}a_2b_1 & C_{22}b_1b_2 \end{pmatrix} \quad (14)$$

Expanding the bilinear form $\mathbf{p}_1^\top \mathbf{M} \mathbf{p}_2$:

$$\begin{aligned} \mathbf{p}_1^\top \mathbf{M} \mathbf{p}_2 &= M_{11} \cos E_1 \cos E_2 - M_{11}e_2 \cos E_1 - M_{11}e_1 \cos E_2 + M_{11}e_1e_2 \\ &\quad + M_{12} \cos E_1 \sin E_2 - M_{12}e_1 \sin E_2 \\ &\quad + M_{21} \sin E_1 \cos E_2 - M_{21}e_2 \sin E_1 \\ &\quad + M_{22} \sin E_1 \sin E_2 \end{aligned} \quad (15)$$

1.5.3. Full constraint

Substituting Equation 11 and Equation 15 into Equation 10 and collecting by trigonometric structure:

$$\begin{aligned} &\underbrace{a_1^2 + a_2^2 - M_{11}e_1e_2 - r_{\text{SOI}}^2}_{\text{constant}} \\ &+ \underbrace{(-2a_1^2e_1 + M_{11}e_2)}_{\alpha_1} \cos E_1 + \underbrace{(-2a_2^2e_2 + M_{11}e_1)}_{\alpha_2} \cos E_2 \\ &+ \underbrace{M_{21}e_2}_{\beta_1} \sin E_1 + \underbrace{M_{12}e_1}_{\beta_2} \sin E_2 \\ &+ a_1^2e_1^2 \cos^2 E_1 + a_2^2e_2^2 \cos^2 E_2 \\ &- M_{11} \cos E_1 \cos E_2 - M_{12} \cos E_1 \sin E_2 \\ &- M_{21} \sin E_1 \cos E_2 - M_{22} \sin E_1 \sin E_2 \\ &= 0 \end{aligned} \quad (16)$$

1.6. Sum-to-Product Reduction

Define sum and difference anomalies $S := E_1 + E_2$, $D := E_1 - E_2$.

1.6.1. Squared cosines

Using $\cos^2 E_i = \frac{1}{2}(1 + \cos 2E_i)$:

$$a_1^2e_1^2 \cos^2 E_1 + a_2^2e_2^2 \cos^2 E_2 = \frac{a_1^2e_1^2 + a_2^2e_2^2}{2} + \frac{a_1^2e_1^2}{2} \cos 2E_1 + \frac{a_2^2e_2^2}{2} \cos 2E_2 \quad (17)$$

1.6.2. Mixed products

Applying the product-to-sum identities $\cos A \cos B = \frac{1}{2}[\cos(A - B) + \cos(A + B)]$, $\sin A \sin B = \frac{1}{2}[\cos(A - B) - \cos(A + B)]$, $\sin A \cos B = \frac{1}{2}[\sin(A + B) + \sin(A - B)]$, $\cos A \sin B = \frac{1}{2}[\sin(A + B) - \sin(A - B)]$:

$$\begin{aligned}
-M_{11} \cos E_1 \cos E_2 &= -\frac{M_{11}}{2}(\cos D + \cos S) \\
-M_{22} \sin E_1 \sin E_2 &= -\frac{M_{22}}{2}(\cos D - \cos S) \\
-M_{21} \sin E_1 \cos E_2 &= -\frac{M_{21}}{2}(\sin S + \sin D) \\
-M_{12} \cos E_1 \sin E_2 &= -\frac{M_{12}}{2}(\sin S - \sin D)
\end{aligned} \tag{18}$$

Collecting coefficients of $\cos D$, $\cos S$, $\sin D$, $\sin S$:

$$-\frac{M_{11} + M_{22}}{2} \cos D - \frac{M_{11} - M_{22}}{2} \cos S + \frac{M_{12} - M_{21}}{2} \sin D - \frac{M_{12} + M_{21}}{2} \sin S \tag{19}$$

1.6.3. Collected constraint

Absorbing the constant $\frac{1}{2}(a_1^2 e_1^2 + a_2^2 e_2^2)$ into the constant term and defining:

$$K := a_1^2 + a_2^2 + \frac{a_1^2 e_1^2 + a_2^2 e_2^2}{2} - M_{11} e_1 e_2 - r_{\text{SOI}}^2 \tag{20}$$

the full constraint becomes:

$$\begin{aligned}
&\alpha_1 \cos E_1 + \alpha_2 \cos E_2 + \beta_1 \sin E_1 + \beta_2 \sin E_2 \\
&+ \frac{a_1^2 e_1^2}{2} \cos 2E_1 + \frac{a_2^2 e_2^2}{2} \cos 2E_2 \\
&- \frac{M_{11} + M_{22}}{2} \cos D - \frac{M_{11} - M_{22}}{2} \cos S \\
&+ \frac{M_{12} - M_{21}}{2} \sin D - \frac{M_{12} + M_{21}}{2} \sin S \\
&+ K \\
&= 0
\end{aligned} \tag{21}$$

where $\alpha_1 = -2a_1^2 e_1 + M_{11} e_2$, $\alpha_2 = -2a_2^2 e_2 + M_{11} e_1$, $\beta_1 = M_{21} e_2$, $\beta_2 = M_{12} e_1$.

1.7. Harmonic Addition

The terms $\alpha_1 \cos E_1 + \beta_1 \sin E_1$ in Equation 21 have the same frequency and can be combined via the harmonic addition theorem:

$$\alpha_1 \cos E_1 + \beta_1 \sin E_1 = R_1 \cos(E_1 - \varphi_1) \tag{22}$$

where:

$$R_1 := \sqrt{\alpha_1^2 + \beta_1^2}, \quad \varphi_1 := \text{atan2}(\beta_1, \alpha_1) \tag{23}$$

Similarly for E_2 , D , and S :

$$\begin{aligned}
&\alpha_2 \cos E_2 + \beta_2 \sin E_2 = R_2 \cos(E_2 - \varphi_2) \\
&- \frac{M_{11} + M_{22}}{2} \cos D + \frac{M_{12} - M_{21}}{2} \sin D = R_D \cos(D - \varphi_D) \\
&- \frac{M_{11} - M_{22}}{2} \cos S - \frac{M_{12} + M_{21}}{2} \sin S = R_S \cos(S - \varphi_S)
\end{aligned} \tag{24}$$

where:

$$R_2 := \sqrt{\alpha_2^2 + \beta_2^2}, \quad \varphi_2 := \text{atan2}(\beta_2, \alpha_2) \tag{25}$$

$$R_D := \frac{1}{2} \sqrt{(M_{11} + M_{22})^2 + (M_{12} - M_{21})^2}, \quad \varphi_D := \text{atan2}(M_{12} - M_{21}, -(M_{11} + M_{22})) \tag{26}$$

$$R_S := \frac{1}{2} \sqrt{(M_{11} - M_{22})^2 + (M_{12} + M_{21})^2}, \quad \varphi_S := \text{atan2}(-(M_{12} + M_{21}), -(M_{11} - M_{22})) \tag{27}$$

Substituting back into Equation 21:

$$R_1 \cos(E_1 - \varphi_1) + R_2 \cos(E_2 - \varphi_2) + \frac{a_1^2 e_1^2}{2} \cos 2E_1 + \frac{a_2^2 e_2^2}{2} \cos 2E_2 + R_D \cos(D - \varphi_D) + R_S \cos(S - \varphi_S) + K \quad (28)$$

1.8. Existence Bounds

1.8.1. Naive bound

Since each cosine lies in $[-1, 1]$, the function $f(E_1, E_2)$ in Equation 28 is bounded in $[K - A, K + A]$ where:

$$A := R_1 + R_2 + \frac{a_1^2 e_1^2}{2} + \frac{a_2^2 e_2^2}{2} + R_D + R_S \quad (29)$$

A necessary condition for solutions is $|K| \leq A$. Equivalently, if $|K| > A$ then **no encounter is possible**. This treats all six cosines as independent, but they share only two degrees of freedom.

1.8.2. Separable decomposition

Split f into terms by their variable dependence:

$$f(E_1, E_2) = g_1(E_1) + g_2(E_2) + h(E_1, E_2) + K \quad (30)$$

where:

$$g_i(E_i) := R_i \cos(E_i - \varphi_i) + \frac{a_i^2 e_i^2}{2} \cos 2E_i \quad (31)$$

$$h(E_1, E_2) := R_D \cos(D - \varphi_D) + R_S \cos(S - \varphi_S) \quad (32)$$

1.8.2.1. Bounding h : amplitude depends on E_1

Since both terms of h have frequency 1 in E_2 (one via $D = E_1 - E_2$, one via $S = E_1 + E_2$), we can collect them. Expanding into $\cos E_2$ and $\sin E_2$ components:

$$h = P(E_1) \cos E_2 + Q(E_1) \sin E_2 \quad (33)$$

where:

$$\begin{aligned} P(E_1) &= R_D \cos(E_1 - \varphi_D) + R_S \cos(E_1 - \varphi_S) \\ Q(E_1) &= R_D \sin(E_1 - \varphi_D) - R_S \sin(E_1 - \varphi_S) \end{aligned} \quad (34)$$

By harmonic addition in E_2 , for any fixed E_1 :

$$|h| \leq \sqrt{P^2 + Q^2} =: \rho(E_1) \quad (35)$$

Computing $P^2 + Q^2$ and using $\cos A \cos B - \sin A \sin B = \cos(A + B)$:

$$\rho(E_1)^2 = R_D^2 + R_S^2 + 2R_D R_S \cos(2E_1 - \varphi_D - \varphi_S) \quad (36)$$

This varies between $(R_D - R_S)^2$ and $(R_D + R_S)^2$, so:

$$|R_D - R_S| \leq \rho(E_1) \leq R_D + R_S \quad (37)$$

The naive bound uses $\rho = R_D + R_S$ everywhere. The tighter bound recognizes that for most values of E_1 , the mixed term has a smaller amplitude.

1.8.2.2. Bounding g_i : coupled harmonics

Each g_i is a sum of first and second harmonics of the same variable. Writing $\theta_i = E_i - \varphi_i$:

$$g_i = R_i \cos \theta_i + \frac{a_i^2 e_i^2}{2} \cos(2\theta_i + 2\varphi_i) \quad (38)$$

The extreme values $g_{i, \min}$ and $g_{i, \max}$ can be found by solving the 1D stationarity condition $g'_i = 0$, which is a cubic in $\cos \theta_i$ — solvable in closed form or cheaply by 1D numerical sweep. The naive bound $|g_i| \leq R_i + a_i^2 e_i^2 / 2$ is not tight because the two harmonics cannot simultaneously achieve their extremes.

1.8.3. Tighter necessary condition

Combining the exact ranges of g_i with the E_1 -dependent bound on h :

$$g_{1,\min} + g_{2,\min} - (R_D + R_S) + K \leq f \leq g_{1,\max} + g_{2,\max} + (R_D + R_S) + K \quad (39)$$

No encounter is possible if the interval does not contain zero:

$$g_{1,\max} + g_{2,\max} + (R_D + R_S) + K < 0 \quad \text{or} \quad g_{1,\min} + g_{2,\min} - (R_D + R_S) + K > 0 \quad (40)$$

For an even tighter check, one can sweep E_1 and use the E_1 -dependent amplitude $\rho(E_1)$ from Equation 36 together with the exact $g_1(E_1)$:

$$\min_{E_2} f = g_1(E_1) + g_{2,\min} - \rho(E_1) + K \quad (41)$$

$$\max_{E_2} f = g_1(E_1) + g_{2,\max} + \rho(E_1) + K \quad (42)$$

If $g_1(E_1) + g_{2,\max} + \rho(E_1) + K < 0$ for all E_1 , no encounter exists. This reduces the 2D problem to a 1D sweep.